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System, apparatus, and method for creating a lumen

Abstract

A method of performing angioplasty includes accessing an artery, inserting a tubular balloon into to the artery in a low-profile operating mode, the tubular balloon constrained in a generally helical shape, and forming a lumen within the generally helical shape by expanding the tubular balloon into a high-profile operating mode. A system for creating a lumen is also disclosed.

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Background/Summary

RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 15/881,280, filed Jan. 26, 2018, which is hereby incorporated by reference in its entirety; which is a divisional of U.S. patent application Ser. No. 15/445,234, filed Feb. 28, 2017, which is hereby incorporated by reference in its entirety and which claims priority to U.S. patent application Ser. No. 14/524,834, filed Oct. 27, 2014; which is hereby incorporated by reference in its entirety, and which claims priority to the following provisional patent applications which are hereby incorporated by reference in their entirety: U.S. Ser. No. 61/896,052 that was filed on Oct. 26, 2013; and U.S. Ser. No. 61/973,510 that was filed on Apr. 1, 2014.

BACKGROUND OF THE INVENTION

(1) The invention is a system, apparatus and method for creating a space (collectively the “system”). More specifically, the system creates a lumen within a body to facilitate the use of a medical device, such as the use of a catheter in a blood vessel. The term “lumen” means a “canal, duct, or cavity of a tubular organ.” Although the system can be implemented in a wide variety of different contexts, the original inspiration for the conceptualization of the system arose in the context of catheterization in the blood vessels of human beings. The system can facilitate catheterization by creating additional “working space” (i.e. the lumen) at a desired location within the body of a patient. The additional space can be created by transitioning from a low-profile operating mode into a high-profile operating mode.

I. Catheterization Procedures

(2) The term “catheter” refers collectively to a wide range of medical devices that are inserted into the body to (1) diagnose a medical condition; (2) treat a medical condition; (3) deliver nourishment; or (4) deliver medicine. The term “catheter” is often used more specifically to refer to a tube inserted into the body of a patient for the purposes of (a) removing material from a location in the body of a patient and/or (b) delivering medicinal and/or nourishing material to a specific location within the body of a patient. Catheters can be used in a variety of locations for a variety of purposes within the body of a patient. Catheterization procedures are commonly involved in the diagnosis and treatment of the cardiovascular system, the excretory system, and other similar systems of a patient.

II. Cardiovascular Disease is a Global Threat

(3) The circulation of blood is essential for a healthy body. Blood provides organs and individual cells with oxygen and nutrients necessary to sustain life. Blood also removes cellular metabolic waste products from the body. The proper flow of blood is a prerequisite for good health. At the center of the cardiovascular system is the heart, an organ responsible for pushing blood throughout

the body. The heart functions as a pump at the center of a complex network of arteries and veins that make up the cardiovascular system. The cardiovascular system is thus responsible for the delivery of oxygen and nutrients and the removal of certain wastes throughout the body. The performance of the cardiovascular system can be evaluated in terms of cardiac output.

(4) Unfortunately, age, disease, trauma, and/or other ailments can hinder the distribution of blood throughout the body. Cardiovascular diseases are a serious health problem in the United States and elsewhere. About 1 in 3 deaths in the US is attributed to cardiovascular disease, which includes heart attacks and strokes. According to the World Health Organization (“WHO”), cardiovascular diseases are the number one cause of death in world. An estimated 17.3 million people died of cardiovascular diseases in 2008, a number that represents 30% of all deaths occurring in that year. According to WHO estimates, the number of deaths caused by cardiovascular diseases will reach 23.4 million by 2030.

(5) The Centers for Disease Control and Prevention (“CDC”) report that “‘cardiovascular disease is the leading killer in every racial and ethnic group in America.’” Many health problems in the United States are either rooted in or manifested as cardiovascular disease. The most common type of heart disease in the United States is coronary artery disease (“CAD”). CAD occurs when plaque builds up in the arteries that supply blood to the heart. This can cause the arteries to narrow over time in a process called atherosclerosis. Plaque buildup can also cause chest pain or discomfort resulting from the inadequate supply of blood to the heart muscle. This is commonly referred to as a condition known as angina. Over time CAD can lead to an irregular heartbeat, a condition known as arrhythmia, and even heart failure.

III. Cardiovascular Catheterization Procedures

(6) A variety of catheterization procedures are used in the prior art to diagnose and treat arterial disease. In the context of cardiovascular disease, a catheter is often a long, thin, flexible, hollow intravascular tube used to access the cardiovascular system of the body. Catheterization is most commonly conducted through the radial artery in the wrist (transradial catheterization) or the femoral artery of the groin (transfemoral catheterization). Catheterization can also be conducted through the elbow, neck, and other parts of the body.

(7) A wide variety of intravascular procedures can be used to address cardiovascular health issues in human beings. Percutaneous coronary intervention (“PCI”) procedures are a type of intravascular procedure commonly referred to as “coronary angioplasty”, “balloon angioplasty” or simply “angioplasty”. Patients suffering from atherosclerosis have narrowed or blocked coronary artery segments resulting from the buildup of cholesterol-laden plaque. Angioplasty is a medical procedure used to treat the narrowed coronary arteries of the heart.

(8) During angioplasty, a cardiologist feeds a deflated balloon or other similar device to the site of the blockage. The balloon can then be inflated at the point of blockage to open the artery. A stent is often permanently placed at the site of blockage to keep the artery open after the balloon is deflated and removed. Angioplasty has proven to be a particularly effective treatment for patients with medically refractory myocardial ischemia. Unfortunately, it is not always possible to position the catheter in the desired location for the purposes of an angioplasty procedure.

IV. Problem of Access

(9) Catheterization procedures can provide a valuable, effective, and minimally invasive option for diagnosing and treating cardiovascular problems and other types of medical problems.

Unfortunately, it is not always possible for prior art tools and techniques to reach the blockage site with a catheter. Blockage within a blood vessel can block catheters as well as blood flow. Two common problems of access are vessel tortuosity and insignificant stenoses. The vessel pathway to the blockage that needs treatment may be very tortuous, which means it is very curved or serpentine and the angioplasty balloon catheter cannot be inserted through the tortuous vessel. Also, a portion of the vessel may be stenosed, which means there are smaller blockages that make the vessel too narrow and prevent insertion of the balloon catheter. These smaller blockages are

usually not intended to be treated with balloon angioplasty. It would be desirable to empower health care providers with enhanced tools and methodologies for working around obstacles to the blockage site.

SUMMARY OF THE INVENTION

(10) In one example, a method of performing angioplasty includes accessing an artery, inserting a tubular balloon into the artery in a low-profile operating mode, the tubular balloon constrained in a generally helical shape, and forming a lumen within the generally helical shape by expanding the tubular balloon into a high-profile operating mode.

(11) In one example, a system for forming a lumen includes a tubular balloon operable in a low-profile operating mode and a high-profile operating mode, and a matrix constraining the tubular balloon in a generally helical shape, the matrix comprising a weave. The tubular balloon has a first diameter in a low-profile operating mode and a second diameter in a high-profile operating mode, and the second diameter is larger than the first diameter.

(12) In another one example, a system for forming a lumen includes a tubular balloon, the tubular balloon operable in a low-profile operating mode and a high-profile operating mode, and a matrix constraining the tubular balloon in a generally helical shape, the matrix comprising at least one thermally formed connection. The tubular balloon has a first diameter in a low-profile operating mode and a second diameter in a high-profile operating mode, and the second diameter is larger than the first diameter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Many features and inventive aspects of the system, are illustrated in the following drawings. However, no patent application can disclose all of the potential embodiments of an invention. In accordance with the provisions of the patent statutes, the principles and modes of operation of the system are explained and illustrated in certain preferred embodiments. However, it must be understood that the system may be practiced otherwise than is specifically explained and illustrated without departing from its spirit or scope.

(2) The description of the system and the various illustrations of the system should be understood to include all novel and non-obvious combination of elements described herein, and claims may be presented in this or a later application to any novel and non-obvious combination of these elements. Moreover, the foregoing embodiments are illustrative, and no single feature or element is essential to all possible combinations that may be claimed in this or a later application.

(3) FIG. 1a is a block diagram illustrating an example of a system for creating a lumen.

(4) FIG. 1b is a flow chart diagram illustrating an example of a process for creating a lumen.

(5) FIG. 1c is an environmental diagram illustrating an example of an expansion component in a low-profile operating mode.

(6) FIG. 1d is an environmental diagram illustrating an example of an expansion component in a high-profile operating mode.

(7) FIG. 2a is a hierarchy diagram illustrating an example of different embodiments of the system, including direct expansion embodiments and indirect expansion embodiments of the system.

(8) FIG. 2b is a hierarchy diagram illustrating an example of different embodiments of the system, including expansion component balloon embodiments and expansion component non-balloon embodiments.

(9) FIG. 2c is a hierarchy diagram illustrating an example of different types of balloons that can be utilized by the system.

(10) FIG. 3a is diagram illustrating a partial and close-up view of the tubular balloon expansion component illustrated in FIG. 3b.

- (11) FIG. 3*b* is a diagram illustrating an example of an axial view of the tubular balloon expansion component.
- (12) FIG. 3*c* is a diagram illustrating an example of a top view of the tubular balloon expansion component.
- (13) FIG. 3*d* is a diagram illustrating an example of a side view of the tubular balloon expansion component.
- (14) FIG. 3*e* is a diagram illustrating an example of a cross-sectional view of a side view of the tubular balloon expansion component with an illustration of a space within the tubular balloon expansion component.
- (15) FIG. 3*f* is a diagram illustrating an example of a partial and close-up view of the tubular balloon expansion component illustrated in FIG. 3*e*.
- (16) FIG. 3*g* is a perspective and partial diagram illustrating an example of a tubular balloon expansion component.
- (17) FIG. 3*h* is a diagram illustrating an example of a front view of a pleated tubular expansion component, an example of a passive expansion component.
- (18) FIG. 3*i* is a diagram illustrating an example of a perspective view of tubular balloon expansion component
- (19) FIG. 4*a* is a flow chart diagram illustrating an example of a process for creating a lumen using a guide balloon embodiment of the system.
- (20) FIG. 4*b* is an environmental diagram illustrating an example of a process step where the guide balloon is inserted.
- (21) FIG. 4*c* is an environmental diagram illustrating an example of a process step where the guide balloon is inflated.
- (22) FIG. 4*d* is an environmental diagram illustrating an example of a process step where the expansion component in the form of a cover is advanced over the inflated guide balloon in order to expand the cover from a low-profile state into a high-profile state.
- (23) FIG. 4*e* is an environmental diagram illustrating an example of a process step where the cover is positioned as desired within the body of the patient to create a lumen at the desired location.
- (24) FIG. 4*f* is an environmental diagram illustrating an example of a process step where the guide balloon is deflated and removed, creating a lumen within the cover.
- (25) FIG. 4*g* is an environmental diagram illustrating an example of a process step where a stent catheter is inserted through the space created by the cover.
- (26) FIG. 5*a* is a flow chart diagram illustrating an example of a process for creating a lumen using an insertion component embodiment of the system.
- (27) FIG. 5*b* is an environmental diagram illustrating an example of a process step where the cover is inserted into the body of the patient.
- (28) FIG. 5*c* is an environmental diagram illustrating an example of a process step where an insertion component is inserted into the cover (a type of expansion component) positioned within the body of the patient to expand the distal section of the expansion component and to create the desired lumen at the desired location.
- (29) FIG. 5*d* is an environmental diagram illustrating an example of a process step where a stent catheter is inserted through the cover.
- (30) FIG. 6*a* is a flow chart diagram illustrating an example of a process for creating a lumen using a sheathed balloon embodiment of the system.
- (31) FIG. 6*b* is an environmental diagram illustrating an example of a process step where a sheath covers the sheathed balloon during insertion the sheathed balloon.
- (32) FIG. 6*c* is an environmental diagram illustrating an example of a process step where the sheath and the sheathed balloon within the sheath are positioned as desired within the body of the patient.
- (33) FIG. 6*d* is an environmental diagram illustrating an example of a process step where the sheath is withdrawn. This causes the balloon to self-expand because it is no longer constrained by

the sheath, triggering the creation of the additional working space (i.e. lumen) within in the body of the patient.

(34) FIG. 6e is an environmental diagram illustrating an example of how the expanded sheathed balloon can create or enhance the lumen at the desired location within the body of the patient.

(35) FIG. 6f is an environmental diagram illustrating an example of a process step where the stent catheter is inserted into the patient through the working space created by the presence of the balloon in a high-profile operating mode.

(36) FIG. 6g is an environmental diagram illustrating an example of a process step where the sheath is advanced to collapse the balloon for removal.

(37) FIG. 7a is a diagram illustrating a perspective view of a helix and matrix configuration that includes a tubular balloon constrained in the shape of a helix by a weave functioning as a matrix.

(38) FIG. 7b is a diagram illustrating an example of a side view of the helix and matrix configuration of FIG. 7a.

(39) FIG. 7c is a diagram illustrating an example of an axial view of the helix and matrix configuration of FIGS. 7a and 7b.

(40) FIG. 7d is a diagram illustrating an example of a perspective section view of the helix and matrix configuration of FIGS. 7a-7c.

(41) FIG. 7e is a diagram illustrating an example of close-up view of the illustration in FIG. 7d.

(42) FIG. 7f is a hierarchy diagram illustrating an example of different components and component configurations that can be utilized in a helix balloon embodiment of the system.

DETAILED DESCRIPTION

(43) The invention is a system, apparatus and method for creating a space (collectively the “system”). More specifically, the system creates a lumen within a body to facilitate the use of a medical device, such as the use of a catheter in a blood vessel. The term “lumen” means a “canal, duct, or cavity of a tubular organ.” Although the system can be implemented in a wide variety of different contexts, the original inspiration for the conceptualization of the system arose in the context of catheterization in the blood vessels of human beings. The system can facilitate catheterization by creating additional “working space” (i.e. the lumen) at a desired location within the body of a patient. The additional space can be created by transitioning from a low-profile operating mode into a high-profile operating mode. The additional space can enable the use of other medical devices by overcoming the problems of conventional access such as vessel tortuosity or insignificant stenoses. The system enables a balloon angioplasty catheter or stent catheter can be inserted through the passageway or tunnel of the lumen past the access problems and onto the desired location.

(44) All of the numbered elements illustrated in the drawings and discussed in the text below that pertain to structural components rather than process steps are defined in the glossary provided in Table 1 below.

I. OVERVIEW

(45) The system can create a lumen in the body of a patient. That lumen can be used to position a medical device, such as a catheter, that can potentially save the life of the patient. The system can be described in terms of interacting entities, components, operational attributes, and processes.

(46) A. Entities

(47) As illustrated in FIG. 1a, a system **100** is an interface between a healthcare provider **92** and a body of a living organism, i.e. a patient **90**. The provider **92** is typically a physician, although nurses, paramedics, physician assistants, veterinarians, and other health care professionals can potentially act as providers **92** in certain contexts. The patient **90** is typically a human being, but other organisms can constitute patients **90** in certain contexts. The system **100** is a tool that the provider **92** can use to benefit the health status of the patient **90**.

(48) B. System

(49) The purpose of the system **100** is to create “working space” (i.e. a lumen **120**) within the body

of the patient **90** sufficient to enable the positioning and use of a medical device **80** such as a catheter within the body of the patient **90**. The system **100** can be implemented in a wide variety of different ways. The system **100** can be used to improve the health of the patient **90** and to even save the life of the patient **90**.

(50) C. Medical Devices and Medical Procedures

(51) A wide variety of different medical devices **80** and medical procedures **81** can benefit from the lumen **120** created by the system **100**. Examples of potentially useful medical devices **80** include but are not limited to all types of catheters, stents, patient monitoring applications, and other similar invasive devices.

(52) A catheter device is potentially any device inserted into the body of a patient **90**. The term “catheter device” refers collectively to a wide range of medical devices that are inserted into the body to (1) diagnose a medical condition; (2) treat a medical condition; (3) delivery nourishment; or (4) deliver medicine. The term “catheter device” is often used more specifically to refer to a tube inserted into the body of a patient **90** for the purposes of (a) removing material from a location in the body of a patient **90** and/or (b) delivering medicinal and/or nourishing material to a specific location within the body of a patient **90**. Catheters can be used in a variety of locations for a variety of purposes within the body of the patient **90**. Catheterization procedures are commonly involved in the diagnosis and treatment of the cardiovascular system, the excretory system, and other systems of a patient **90**.

(53) The system **100** was originally conceived for the purpose of serving providers **92** involved in providing medical procedures **81** such as coronary vascular procedures. Examples of such procedures include but are not limited to Percutaneous Coronary Intervention (PCI), Percutaneous Coronary Angiogram (PCA), Chronic Total Occlusions (CTO), Stent implantation, Atherectomy, and Embolic Protection. The system **100** can be particularly useful in the context of transradial catheterizations (catheterizations in which the catheter initially enters the body of the patient **90** through the radial artery) because transradial catheterizations typically involve catheterization devices with a relatively smaller profile and relatively sparse space in which to operate. The system **100** in its varying embodiments can also be used in a variety of contexts that involve cardiovascular care and the treatment of wholly different conditions.

(54) D. Lumen

(55) A lumen **120** is a space created within the patient **90** by the system **100**. The lumen **120** is often referred to as a “canal, duct, or cavity within a tubular organ”. The lumen **120** is the “working space” within the patient **90** in which the medical device **80** is positioned. In many embodiments of the system **100**, the lumen **120** is located within the expansion component **110** and the expansion component **110** is at least substantially in the form a hollow tube, with the lumen **120** comprising the hollow core of the expansion component **110**.

(56) E. Expansion Component

(57) An expansion component **110** is the device capable of existing in at least two operating modes **130**, a low-profile operating mode **132** and a high-profile operating mode **134**.

(58) There are a wide variety of different embodiments of expansion components **110** that can be incorporated into a wide variety of different embodiments of the system **100**. In many embodiments of the system **100**, the expansion component **110** can transform from a high-profile operating mode **134** back into a low-profile operating mode **132** when the expansion component **110** is no longer needed. In many embodiments, it will be easier for the provider **92** to remove the expansion component **110** from the patient **90** when the expansion component **110** is in a low-profile operating mode **132**.

(59) Expansion components **110** can be categorized as direct vs. indirect. Some embodiments of the system **100** utilize balloons as expansion components **110** while other embodiments of the system **100** utilize non-balloon expansion components **110**.

(60) F. Operating Modes/States

(61) The expansion component **110** can operate in two or more operating modes **130** (which can also be referred to as states **130**). The low-profile operating mode **132** is typically the most convenient operating mode **130** in which to insert the expansion component **110** into the patient **90** prior to creating the lumen **120**. The low-profile operating mode **132** is also typically the most convenient operating mode **130** in which the provider **92** can remove the expansion component **110** after the lumen **120** is created and after the medical device **80** has been positioned correctly within the patient **90**.

(62) Some embodiments of the system **100** will involve one or more intermediate operating modes between the low-profile operating mode **132** and the high-profile operating mode **134**.

(63) G. Process Flow View

(64) The system **100** can be described as a series of process steps as well as a configuration of interacting elements. FIG. **1b** is a flow chart diagram illustrating an example of a method for creating a lumen **120**.

(65) At **200**, the expansion component **110** is inserted within the patient **90**. Different embodiments of the system **100** can involve different types of expansion components **110** to create lumen **120** for different types of medical devices **80**.

(66) At **202**, the expansion component **110** is positioned within the patient **90**. Different embodiments of the system **100** can involve a wide variety of different locations within the body of the patient **90**.

(67) At **204**, the operating mode **130** of the expansion component **110** is changed from a low-profile operating mode **132** into a high-profile operating mode **134** in order to create a lumen **120**. It is the lumen **120** that serves as the “working space” for the proper positioning and use of the medical device **80**, such as a catheter.

(68) In many embodiments, after the lumen **120** is created and medical device **80** is properly positioned, the expansion component **110** is transformed back from a high-profile operating mode **134** into a low-profile operating mode **132** to facilitate the removal of the expansion component **110** from the body of the patient **90**.

(69) H. Operating Environment

(70) The system **100** can be implemented in a wide variety of different operating environments and locations. The process of determining which embodiment of the system **100** is best suited for a particular context should begin with identifying the desired medical device **80** to be used at the desired location. The appropriate expansion component **110** can then be identified and selected.

(71) FIG. **1c** is an environmental diagram illustrating an example of an expansion component **110** in a low-profile operating mode **132**. The expansion component **110** is being positioned to a desired location **88** within a blood vessel **91** in the patient **90**.

(72) FIG. **1d** is an environmental diagram illustrating an example of an expansion component **110** that has been transformed (i.e. expanded) from a low-profile operating mode **132** into a high-profile operating mode **134**.

(73) I. Ancillary Components

(74) In many embodiments of the system **100**, the expansion component **110** is but one component of many. For example, in the illustrations of FIGS. **1c** and **1d** the expansion component **110** can interface with certain ancillary components, such as a guide catheter **121** and a guide wire **122**. In navigating the various narrow blood vessels **91** a variety of guide catheters **121** and guide wires **122** may be utilized to position the expansion component **110** to the desired location **88**. Such components may be part of the system **100**, but the use of ancillary components will vary widely between different embodiments of the system **100**. The system **100** can include virtually any prior art component useful to the provider **92** in addressing the needs of the patient **90**.

II. ALTERNATIVE EMBODIMENTS

(75) Many features and inventive aspects of the system **100** are illustrated in the figures and described in the text of this application. However, no patent application can disclose all of the

potential embodiments of an invention. In accordance with the provisions of the patent statutes, the principles and modes of operation of the system **100** are explained and illustrated in certain preferred embodiments. However, it must be understood that the system **100** may be practiced otherwise than is specifically explained and illustrated without departing from its spirit or scope. (76) The description of the system **100** and the various illustrations of the system **100** should be understood to include all novel and non-obvious combination of elements described herein, and claims may be presented in this or a later application to any novel and non-obvious combination of these elements. Moreover, the foregoing embodiments are illustrative, and no single feature or element is essential to all possible combinations that may be claimed in this or a later application. (77) There are various categories that can be useful in describing various embodiments of the system **100**.

A. Direct Vs. Indirect

(78) With respect to all embodiments of the system **100**, the expansion component **110** expands from a low-profile operating mode **132** into a high-profile operating mode **134** to create a lumen **120**. For some embodiments of the expansion component **110**, the transformation between operating modes **130** is accomplished directly by the expansion component **110** while in other embodiments of the expansion component **110**, the transformation between operating modes is accomplished only indirectly by the expansion component **110**.

(79) FIG. **2a** is a hierarchy diagram illustrating examples of direct expansion embodiments **101** as well as indirect expansion embodiments **102**. Indirect expansion embodiments **102** involve expansion components **110** that expand or shrink due to other components of the system **100**. In contrast, direct expansion components **101** involve expansion components **110** that can change operating modes **130** without the use of other components of the system **100**.

(80) Direct expansion embodiments **101** can include but are not limited to a tubular balloon embodiment **103** and a helix balloon embodiment **104**. Direct expansion embodiments **101** typically involve “inflating” a balloon with a substance such as liquid to expand from a low-profile operating mode **132** into a high-profile operating mode **134**. Some embodiments may utilize a gas, but it is often not desirable to risk inserting bubbles of air or other gases in the blood vessels **91** of patients **90**.

(81) Indirect expansion embodiments **102** can include but are not limited to a guide balloon embodiment **105** (where an expansion component **110** in the form of a cover **116** expands by advancing upon an inflated guide balloon **115**), an insertion component embodiment **106** (where an expansion component **110** in the form of a cover **116** expands through the insertion of an insertion component **117** into the expansion component **110**), and a sheath embodiment **107** (where the sheathed balloon **118** inflates when no longer constrained by the sheath **119**). Indirect expansion embodiments **102** utilize other components of the system **100** to “inflate” to a high-profile operating mode **134** and to “deflate” to a low-profile operating mode **132**. Guide balloon embodiments **105** of the system **100** use an expansion component **110** that is advanced over an inflated balloon to expand the expansion component **110**. Insertion component embodiments **106** of the system **100** use a insertion component **117** that is inserted into the expansion component **110** to expand the expansion component **110**. Sheath embodiments **107** utilize a sheath to constrain an expansion component **110** that would otherwise exist in an expanded state.

B. Expansion Component Balloons Vs. Non-Balloons

(82) Just as different embodiments of the system **100** can be categorized on whether the expansion component **110** is directly or indirectly expanded, the various embodiments of the system **100** can also be categorized on the basis of whether the expansion component **110** is some type of balloon (which inflates using air, some other gas, some form of liquid or fluid, or through the use of mechanical means) or whether the expansion component **110** is not a balloon.

(83) FIG. **2b** is a hierarchy diagram illustrating examples of both expansion component balloon embodiments **108** and expansion component non-balloon embodiments **109**.

(84) Examples of expansion component balloon embodiments **108** can include but are not limited to tubular balloon embodiments **103**, helix balloon embodiments **104**, and sheath embodiments **107**.

(85) Examples of expansion component non-balloon embodiments **109** can include but are not limited to guide balloon embodiments **105** and insertion component embodiments **106**.

C. Active Vs. Passive Expansion Components

(86) Many differences in various embodiments of the system **100** are dictated by the differences in the expansion components **110** of the different embodiments. Two overarching categories of expansion components **110** can be differentiated on the basis of whether they are “active” or “passive”.

(87) 1. Active Expansion Components/Active Apparatuses

(88) a. Balloon without Sheath

(89) The embodiment of the system **100** illustrated in FIGS. **3a-3g** involves an inflatable balloon as the expansion component **110**. That embodiment of the system **100** has a balloon as the expansion component **110** that can be in either a low-profile state **132** or a high-profile state **134** (i.e. an expanded state). The system **100** is transitioned between states **130** by inflating or deflating the expansion component **110** (i.e. the balloon). The system **100** has an “active” control through the inflation and deflation feature.

(90) b. Balloon with Sheath

(91) An alternate embodiment of an active control system **100** is a self-expanding balloon with a sheathed balloon **118** as the expansion component **110**. The system **100** would have a balloon that self-expands. Active control of the system **100** is through the use of a sheath **119** that covers the balloon. The device is in the low-profile state **132** when the sheath **119** covers the self-expanding balloon. In this state **132** the system **100** can be inserted to the required location. The low-profile state **132** will facilitate insertion in an atraumatic manner. In this state **132**, the system **100** will be able to interface with other necessary devices, such as a 0.014 coronary guide wire and a guide catheter. When the system **100** is properly positioned at the required location, the sheath **119** is retracted by active control which allows the expansion component **110** to self-expand to the expanded high-profile state **134**. In the expanded high-profile state **134** the system **100** can enable the performance of medical procedures **81** involving the insertion of other medical devices **80** such as a catheter device. It will provide a space **120** through which other devices can be inserted. When the expanded state **134** is not required anymore, the sheath **119** can be advanced over the balloon **118** with active control and transition the system **100** back to the low-profile state **132**.

(92) Another potential alternative means to achieve a self-expanding expansion component **110** is to use materials with a spring feature. Many metals have a spring feature, such as stainless steels. Alternately, shape memory metals such as Nitinol could be used to achieve a self-expanding feature. It is envisioned that there may be other materials, either metals or non-metals, which could be used to achieve a self-expanding feature. These materials can be used to make a structure that serves as a “sheathed balloon” **118**. In some embodiments, the sheathed balloon **118** can be similar to other types of balloons **111**. In other embodiments, the sheathed balloon **118** can be a self-expanding braid structure **124**.

(93) 2. Passive Expansion Components/Passive Apparatuses

(94) A passive control system is a system **100** that has two or more operating modes **130**, and the system **100** is passively transitioned between the states **130** instead of actively transitioned between states **130**.

(95) a. Pleated Expansion Component

(96) One embodiment of a passive control is a pleated expansion component **110** as illustrated in FIGS. **3h** and **3i**. The expansion component **110** of the system **100** would be made with pleats. The pleats cause the expansion component **110** to have a low-profile state **132**. The expansion component **110** is small because of its pleated shape. When a different medical device **80** is inserted

into the space **120**, or pleated expansion component **110**, it will passively expand to the larger expanded state **134** to allow the other medical device **80** to pass through. The other medical device **80** will force the pleats to expand outward to form a larger space **120** and a more expanded expansion component **110**. For this embodiment, the system **100** is passively transitioned between the two states **130** by the insertion of the assisted device, not the active operation of the system **100** by the operator.

(97) b. Elastic Expansion Component

(98) An alternate embodiment of a passive control system **100** is an elastic expansion component **110**. The elastic expansion component **110** would be made of elastic or stretchable materials. The expansion component **110** would be made in the low-profile state **132**. Its cross section is likely to be a round shape, but other shapes are possible, such as elliptical. When a different medical device **80** is inserted into the elastic expansion component **110** it will passively expand to a larger state to allow the other medical device to pass through. The other medical device **80** will force the elastic expanding component **110** to form a larger space **120**. For such an embodiment, the system **100** is passively transitioned between the two states **130** instead of actively transitioned by the operator. A system **100** of this design could be made from a variety of materials, such as medical grade silicones or urethanes.

D. Embodiment Categories

(99) As illustrated in both FIG. **2a** and FIG. **2b**, the various embodiments of the system **100** can be organized into categories. As illustrated in FIG. **2c**, many different embodiments of the system **100** can utilize some form of a balloon **111**. Some embodiments of the system **100** can utilize a balloon **111** with a default state of uninflated that require inflation to transition from a low-profile operating mode **132** into a high-profile operating mode **134** (i.e. the tubular balloon **112** and the helix balloon **113**). Other embodiments of the system **100** use the balloon **111** not as the expansion component but as a mechanism for expanding the expansion component **110** from a low-profile operating mode **132** into a high-profile operating mode **134** (i.e. the guide balloon **115** on which a cover **116** is advanced). Still other embodiments utilize a balloon **111** that has a default state of inflated or that self-inflates (i.e. a sheathed balloon **118**). A sheathed balloon **118** transitions from a low-profile operating mode **132** into a high-profile operating mode **134** when it is removed from the constraining sheath **119**. The sheathed balloon **118** can be returned to the low-profile operating mode **132** by being positioned back within the sheath **119**.

(100) The system **100** can be implemented using expansion components **110** that are (1) integrated into a single stand-alone device with other components of the system **100**; (2) a non-integrated collection of components configured to function with certain supporting components; (3) a magnitude of integration that falls between these two polar opposites.

(101) As indicated by the various arrows in FIG. **1a**, the system **100** can directly interact with both the patients **90** and providers **92**. Such a system **100** can be implemented in a wide variety of different alternative embodiments. Some embodiments of the system **100** can be single stand-alone components, such as an expandable balloon **111**. Other embodiments of the system **100** can involve configurations of multiple components which may be permanently attached to each other, or merely configured to temporarily act in concert with each other.

(102) The system **100** can be used in conjunction with virtually any catheter device **80** and as part of virtually any catheterization procedure. It facilitates a catheterization procedure by aiding the insertion of medical devices **80** such as various catheters and potentially other devices to the desired location **80** in the body of the patient **90** that cannot otherwise be reached without the space **120** created by the system **100** transitioning from a low-profile operating environment **132** into a high-profile operating environment **134**.

(103) By way of example, an angioplasty balloon catheter or a stent catheter may not otherwise be able to be placed in the desired location **88** where the blockage is located. The system **100** can facilitate inserting the balloon or stent **123** (i.e. the catheter device) to the blockage.

(104) The advantage of the system **100** is that it can be inserted to required locations by itself that medical devices **80** such as catheters cannot be inserted by themselves. The ability to exist in either of two states **130** enables the system **100** to have this advantage. Unlike medical devices **80** such as catheterization devices that expand to remove blockage in an artery, the system **100** can be configured for the purpose of merely expanding sufficiently to create operating space for the catheter device. The operating space **120** is in the form of a lumen or passageway created by the expanded state of the system **100**. Other catheterization devices can pass through the operating space **120** in order to be inserted to their desired location **88**. The operating space **120** can create safe passage for catheterization devices **88** through tortuous (serpentine) vessels **91** or past stenoses that impinge vessels **91**. The system **100** may temporarily straighten out tortuous vessels or dilate stenosed areas.

(105) The system **100** works in a supportive role with respect to a medical device **80**, such as catheter. In the context of cardiovascular catheterization, the system **100** is typically inserted into coronary arteries, or other arteries or veins (collectively “vessels” **91**). The system **100** can be appropriately sized and constructed to accomplish the desired task of creating an additional space **120** for the desired catheter device at the desired location **88**. The system **100** can have two or more states **130**, with a low-profile state **132** for insertion and removal of the device, and an expanded state **134** for coronary stabilization.

(106) The original context inspiring the conception of the system **100** was to facilitate percutaneous coronary intervention (PCI) procedures, or other similar intravascular procedures. However, the system **100** can be configured for use with virtually any catheter device and any catheterization procedure.

(107) The system **100** can be made from biocompatible medical grade materials, such as polymers (plastics) and metals. The system **100** may be made from materials or have coatings that give it additional features. It may have a hydrophilic feature. It can be made using various manufacturing methods, such as extrusion, injection molding, thermal forming, thermal bonding, wire forming methods, laser manufacturing methods or other manufacturing methods. It will be made in such a way that it can be properly packaged and sterilized. Likely sterilization methods would be e-beam radiation, gamma radiation, ethylene oxide (EO) gas sterilization or nitrous oxide (NO₂) gas sterilization.

(108) 1. Tubular Balloon Embodiments

(109) In a tubular balloon embodiment **103** of the system **100**, the expansion component **110** is a tubular balloon **112**. FIGS. 3a-3i pertain to tubular balloon embodiments **103** of the system **100**.

(110) The tubular balloon **112** can be inflated with air, other forms of gas, water, and other forms of liquids or fluids. In some tubular balloon embodiments **103**, the tubular balloon **112** can be inflated with mechanical means such as a spring that is uncompressed or other similar means.

(111) 2. Helix Balloon Embodiments

(112) In a helix balloon embodiment **104** of the system **100**, the expansion component **110** is a helix balloon **113**, i.e. a tubular balloon **112** that is constrained by a matrix **114** to form an at least substantially helical shape. FIGS. 7a-7e illustrate examples of helix balloon embodiments **104**.

(113) Just as with tubular balloon embodiments **103**, helix balloon embodiments **104** can utilize a wide variety of different inflating mechanisms.

(114) Helix balloon embodiments **104** can be highly desirable because of the impact of the matrix **114**, which can selectively increase the rigidity of the expansion component **110** so that it can be inserted into locations **88** that a tubular balloon **112** without a matrix **114** will not be able to reach. As illustrated in FIG. 2c, helix balloons **113** can be implemented as conventional inflatable balloons, but also as a self-expanding helix component **141** or as a mechanically-expanding helix component **142**.

(115) 3. Sheath Embodiments

(116) A sheath embodiment **107** of the system **100** uses a balloon **111** that does not require inflation

to transition from a low-profile operating mode **132** into a high-profile operating mode **134**. FIGS. **6a-6g** pertain to sheath embodiments **107** of the system **100**. A sheathed balloon **118** transitions from a low-profile operating mode **132** into a high-profile operating mode **134** when it is removed from the constraining sheath **119**. The sheathed balloon **118** can be returned to the low-profile operating mode **132** by being positioned back within the sheath **119**.

(117) As illustrated in FIG. **2c**, a sheathed balloon **118** can be implemented as a braid balloon **124**.

(118) 4. Guide Balloon Embodiments

(119) A guide balloon embodiment **105** of the system **100** involves an expansion component **110** that is not a balloon **111**. Rather, the expansion component **110** is a cover **116** that is advanced over a preceding inflated balloon, i.e. a guide balloon **115**. FIGS. **4a-4g** illustrated examples of guide balloon embodiments **105** of the system **100**.

(120) 5. Insertion Component Embodiments

(121) Insertion component embodiments **106** of the system **100** need not use any kind of balloon **111** in the expansion/shrinkage processes. In an insertion component embodiment **106** of the system **100**, an insertion component **117** is inserted into the expansion component **110** to cause the expansion component **110** to expand from a low-profile operating mode **132** into a high-profile operating mode **134**. The expansion component **110** in an insertion component embodiment **106** of the system **100** can be a cover **116**, such as another catheter. Insertion component embodiments **106** are illustrated in FIGS. **5a-5d**.

III. TUBULAR BALLOON EMBODIMENTS

(122) Some embodiments of the system **100** will utilize a single tubular balloon **112** to serve as the expansion component **110** to facilitate the transition between a low-profile state **132** and a high-profile state **134** that can create a lumen **120** for the applicable medical device **80**, such as a balloon angioplasty catheter or stent **123**, at the desired location **88** in the body of the patient **90**.

(123) The “working space” or lumen **120** created by the expansion of a tubular balloon **112** into a high-profile operating mode **134** is created within the tubular balloon **112**. Examples of different types of expansion components **110** can include inflatable balloons **112** with a “donut hole” space (see FIGS. **3a-3i**),

(124) As discussed above, some embodiments of the system **100** can be configured to expand/contract using different technologies and different component configurations. In some embodiments of the system **100**, the expansion of the system **100** is achieved through an expansion component **110** that is part of the system **100**. In other embodiments, the expansion of the system **100** is achieved by the expansion of a separate component/device in the system **100** that is expanded, and used to then expand or allow for the expansion of the system **100**. For example, the removal of a sheath **119** can trigger the expansion of the sheathed balloon **118** in a sheath embodiment **107** of the system **100** (see FIGS. **6a-6g**).

(125) Tubular balloons **112** can be implemented in a wide variety of different ways. Some embodiments of tubular balloons **112** as expansion components **110** can use an inflation tube **150** connected to a valve **151** on the tubular balloon **112** to inflate the tubular balloon **112**.

(126) Tubular balloons **112** can be inflated using air, other forms of gases, water, and other forms of liquids or fluids. Tubular balloons **112** can also be inflated using mechanical means such as springs. Some embodiments of tubular balloons **112** can involve a balloon **111** that self-inflates.

(127) For tubular balloon embodiments **103** that require active inflation, the valve **151** is typically positioned at the proximal end of the balloon **112**, which would be like the ‘tail’ end of the balloon **112**. The valve **151** is connected to an inflation tube **150**. The tube **150** runs longitudinally to the inflatable lumen **120**. The inflatable lumen is at the distal end, which would be like the ‘business’ end. The overall length is approximately 100-120 cm (39.4-47.2 inches). The inflatable balloon **112** is approximately 35 mm (1.38 inches). The inflation tube **150** is approximately 65-85 cm (25.6-33.5 inches) in some embodiments of the system **100**. The system **100** can be constructed to have a low-profile state **132**, which would be a deflated or collapsed state. The low-profile diameter size

would be small enough to fit into the required arterial locations and to interface with other medical devices **80** used during the procedure. The low-profile diameter size would be approximately 0.030-0.060 inch (0.76-1.52 mm).

(128) FIG. **3a** is a diagram illustrating a partial and close-up view of the system **100** in FIG. **3b**. A partial example of the inflatable balloon **112** is illustrated along with the accompanying lumen **120** and the tube **150** that facilitates inflation/deflation.

(129) FIG. **3b** is a diagram illustrating an example of an axial view of the system **100**. The lumen **120** created by the system **100** is in the form of a “donut hole” at the center of the expansion component **110**.

(130) FIG. **3c** is a diagram illustrating an example of a top view of the system **100**.

(131) FIG. **3d** is a diagram illustrating an example of a side view of the system **100**.

(132) FIG. **3e** is a diagram illustrating an example of a cross-sectional view of a side view of the system **100** with an illustration of a lumen **120** within the system **100**.

(133) FIG. **3f** illustrates a close-up and partial view of FIG. **3e**.

IV. GUIDE BALLOON EMBODIMENTS

(134) Some embodiments of the system **100** anticipate that a guide balloon **115** is used in conjunction with the system **100**. The guide balloon **115** can help position the system **100** within the body of the patient **90**.

(135) FIG. **4a** is a flow chart diagram illustrating an example of a process for enhancing catheterization performed by a guide balloon embodiment **105** of the system **100**.

(136) At **302**, the guide balloon **115** is inserted into the body of the patient **90**. FIG. **4b** is an environmental diagram illustrating an example of a process step where the guide balloon **115** is inserted. At the beginning of a coronary catheterization procedure a guide catheter **121** or similar medical device **80** can be inserted to the femoral or radial artery, and the guide catheter will be advanced until it accesses the right or left coronary ostium. The ostium is the start of the coronary artery. It is where the artery branches off the aorta. A guide wire **122** will be inserted through the guide catheter **121** and into the coronary artery beyond the point where treatment is to be conducted. The guide balloon **115** of the system **100** will be inserted over top of the guide wire **122** and through the guide catheter **121** into the artery. The guide balloon **115** is in a deflated state while it is inserted. It is inserted past any tortuous areas or stenosis.

(137) Returning to FIG. **4a**, at **304** the guide balloon **115** is inflated. FIG. **4c** is an environmental diagram illustrating an example of a process step where the guide balloon **115** is inflated. The guide balloon **115** is inflated after it is properly positioned. It can be inflated pneumatically with a gas such as air or hydraulically with a liquid. It is most likely to be inflated with a 50-50 mixture of sterile saline and contrast media. It may be inflated to lower pressures of 1-4 atmospheres or higher pressures up to 16 atmospheres. The inflated outside diameter of the guide balloon **115** may be less than, equal to, or greater than the diameter of the artery. The guide balloon **115** may temporarily straighten any tortuous areas of the artery, either completely or partially.

(138) Returning to FIG. **4a**, at **306** the cover **116** is advanced over the guide balloon **115**. FIG. **4d** is an environmental diagram illustrating an example of a process step where the cover **116** is advanced over the inflated guide balloon **115**. The expansion component **110**, which is the core component of the system **100**, is inserted over top of the guide balloon **115** and through the guide catheter **121**. In this embodiment of the system **100** the expansion component **110** may be either a self-expanding design or a fixed diameter design. As the expansion component **110** exists the distal end of the guide catheter **121** it will track over top of the inflated guide balloon **115**. The guide balloon **115** outside diameter and the expansion component **110** inside diameter will be specifically designed for an optimum interface. The interface may be a slip fit design, a line-to-line fit design, or an interference design. The interface design will aid insertion of the expansion component **110** and make insertion as atraumatic as possible to eliminate or prevent arterial wall damage.

(139) FIG. **4e** is an environmental diagram illustrating an example of a cover **116** expanded over a

guide balloon **115**. The guide balloon **115** serves the important task to eliminate or prevent arterial wall damage from the leading edge of the expansion component **110** while it is being inserted, even though the leading edge may be design with its own atraumatic tip. To this end, the guide balloon **115** may intentionally be longer than the expansion component **110**. It may be two times or more than the length of the expansion component **110**.

(140) Returning to FIG. **4a**, at **308** the guide balloon **115** is deflated. FIG. **4f** is an environmental diagram illustrating an example of a process step where the guide balloon **115** is deflated and removed. The guide balloon **115** is deflated and removed after the expansion component **110** is properly positioned. The expansion component **110** may be designed to maintain straightening of the artery after the guide balloon **115** is removed.

(141) Returning to FIG. **4a**, at **310** the guide balloon **115** is removed. The expansion component **110** may be either a self-expanding design or a fixed diameter design for this embodiment of the system **100**. The expansion component **110** will create space **120** in the artery in the form of a lumen. Other devices **80** can pass through the space **120** created by the system **100** when it is in the high-profile expanded state **134**, such as an angioplasty balloon, a stent catheter, or some other form of similar medical device **80**.

(142) At **312**, a stent **123** is positioned through the system **100**. FIG. **4g** is an environmental diagram illustrating an example of a process step where a stent **123** is inserted through the space **120** created by the system **100**.

(143) The system **100** is removed from the artery when it is not needed anymore. The artery would regain its natural shape. This embodiment of the system **100** would interface with the other catheterization devices **80** used during the procedure, such as the guide wire **122**, guide catheter **121**, balloon catheters and stent **123**.

V. INSERTION COMPONENT EMBODIMENTS

(144) FIG. **5a** is a flow chart diagram illustrating an example of a process for enhancing catheterization performed by an insertion component embodiment **106** of the system **100**. This embodiment of the system **100** uses an insertion component **117** that is inserted into the expansion component **110** of a cover **116**. In some embodiments, the insertion component **117** can be attached to the guide catheter **121**.

(145) At **322**, the cover **116** attached to the guide catheter **121** is inserted into the body of the patient **90**. FIG. **5b** is an environmental diagram illustrating an example of a process step where the cover **116** is inserted into the body of the patient **90**. At the beginning of a typical coronary catheterization procedure a guide catheter **121** will be inserted to the femoral or radial artery, and the catheter **121** will be advanced until it accesses the right or left coronary ostium. The ostium is the start of the coronary artery. It is where the artery branches off the aorta. A guide wire **122** will be inserted through the guide catheter **121** and into the coronary artery beyond the point where treatment is to be conducted. For this embodiment of the expansion component **110**, which is in the form of a cover **116**, the cover **116** will often be an integral part of the guide catheter **121**. The cover **116** can be connected to the distal end of the guide catheter **121** as part of the manufacturing process for those components.

(146) Returning to FIG. **5a**, at **324** an insertion component **117** is inserted into the cover **116**. FIG. **5c** is an environmental diagram illustrating an example of a process step where an insertion component **117** is inserted into the cover **116** positioned within the body of the patient **90** to expand the distal section of the cover **116**. An insertion component **117** would be inserted into the inside the entire length of the connected expansion component **110** (i.e. the cover **116**) and guide catheter **121**. As it is inserted it will expand the expansion component **110** (i.e. the cover **116**) to the high-profile state **134**.

(147) Returning to FIG. **5a**, at **326** a stent catheter **123** is inserted into the body of the patient **90** through the insertion component **117**. FIG. **5d** is an environmental diagram illustrating an example of a process step at **326**. The nested structure of the high-profile state **134** expansion component

- 110** and the insertion component **117** will create space **120** through which other medical devices **80** can be inserted, such as an angioplasty balloon catheter or a stent catheter **123**.
- (148) The expansion component **110** (i.e. the cover **116**) of the system **100** and insertion component **117** will be removed when they are not needed anymore.
- (149) The expansion component **110** of this embodiment can be made with shape memory materials, a braid construction, a pleated design or any other expandable design structure.
- (150) Shape memory materials can be metallic or non-metallic. Nitinol is one possible metallic material that could be used. The expansion component **110** could be made from Nitinol and the memorized shape would be the low-profile state **132**. This memorized low-profile state **132** would enable the connected expansion component **110** and guide catheter **121** to be inserted into the coronary artery past the ostium, tortuous areas and any stenoses. The insertion component **117** would be used to actively transition the expansion component **110** from the low-profile state **132** to the high-profile state **134**. Non-metallic shape memory polymers could also be used to construct the expansion component **110** and accomplish the same result.
- (151) A braid structure could be used to construct the cover **116**. The braid would be made to the size of the low-profile state **132**. The woven mesh pattern of the braid has space in the interstices between its wires. This would allow it to expand to the high-profile state **134** when the insertion component **117** is inserted.
- (152) A pleated design could be used to make the cover **116**. The pleated design would be made to the size of the low-profile state **132**. The insertion component **117** would unfold the pleats, when it is inserted, allowing it to transition to the high-profile state **134**.

VI. SHEATHED BALLOON EMBODIMENTS

- (153) FIG. **6a** is a flow chart diagram illustrating an example of a process of enhancing catheterization performed by a sheath covered embodiment **107** of the system **100**. In this category of embodiments, expansion component **110** of the system **100** is self-expanding. The sheath **119** allows for the expansion component **110** to exist in a low-profile mode **132** by constraining the expansion component **110**. Once the expansion component **110** is released from the sheath **119**, the expansion component **110** (such as a sheathed balloon **118**) expands into a high-profile operating mode **134**.
- (154) The self-expanding feature can be made with self-expanding materials, such as a braid structure. The braid structure is cylindrical in shape. The wall of the cylinder is constructed of the woven mesh of the braid. The ends of the cylinder are open. The braid would be designed with space in its weave pattern, which would allow the braid structure to exist in either the high-profile self-expanded state **134** or the low-profile state **132**.
- (155) At **350**, the system **100** with sheath **119** (and the encapsulated expansion component **110** such as a sheathed balloon **118**) is inserted into the body of the patient **90**. FIG. **6b** is an environmental diagram illustrating an example of a process step where a sheath **119** covers the system **100** during insertion. The expansion component **110** could be compressed to a low-profile state **132** and inserted into a sheath **119**. The sheath **119** would cover the expansion component **110** keeping it in the low-profile state **132**. The expansion component **110** and sheath **119** would be inserted through the guide catheter **121** and into the artery **91** as one unit.
- (156) Returning to FIG. **6a**, at **352** the system **100** is positioned within the body of the patient **90**. FIG. **6c** is an environmental diagram illustrating an example of a process step where the sheath **119** and system **100** are positioned as desired within the body of the patient **90**. The expansion component **110** and sheath **119** would have an appropriate low-profile size, strength, and flexibility to be inserted past any tortuous areas or stenosis.
- (157) Returning to FIG. **6a**, at **352** the sheath **119** is withdrawn. FIG. **6d** is an environmental diagram illustrating an example of a process step where the sheath **119** is withdrawn; causing the system **100** to self-expand and triggering the creation of the additional working space **120** within in the body of the patient **90** for the purposes of catheterization. The sheath **119** is removed after the

system **100** is properly positioned. The expansion component **110** will automatically deploy because of its self-expanding feature. The expansion component creates space **120** in the artery. (158) Returning to FIG. **6a**, at **354** the system **100** is expanded into a high-profile state **134**. FIG. **6e** is an environmental diagram illustrating an example of how the expanded system **100** can straighten out an artery within the body of the patient **90**. The expansion component **110** may partially or completely straighten any artery tortuosity. The straightening effect would be transient. When the system **100** is withdrawn the artery would regain its natural shape (159) Returning to FIG. **6a**, at **356** the stent catheter **123** is inserted through the system **100**. FIG. **6f** is an environmental diagram illustrating an example of a process step where the stent catheter **123** is inserted into the patient **90** through the working space **120** created by the presence of the system **100** in a high-profile operating mode **134**. Other devices can pass through the space **120** created by the system **100** when it is in the high-profile expanded state **134**, such as an angioplasty balloon catheter or stent **123**. (160) Returning to FIG. **6a**, at **358** the sheath **119** is advanced to collapse the system **100** for removal. FIG. **6g** is an environmental diagram illustrating an example of a process step where the sheath **119** is advance to collapse the system **100** for removal. The system **100** can be removed when it is not needed any more. The sheath **119** is advanced over the expansion component **110** causing it to collapse to the low-profile state **132**, and then the expansion component **110** and sheath **119** are removed together as one unit. (161) An alternate embodiment of this form of the system **100** uses a self-expanding braid structure **124** to serve as the sheathed balloon **118**. The construction of the braid **124** can be designed to provide optimum performance. Braid **124** characteristics such as number of wires, shape of wire, wire material, pitch, uniform pitch, variable pitch and weave pattern can be chosen to obtain the desired performance. More or less wires, and wire material, can affect strength and flexibility of the component. Round wires or flat wires can affect wall thickness. Pitch and weave pattern can affect expansion strength and profile size. (162) Stainless Steel or Nitinol are likely materials for the braid **124** wire, however other metals or non-metals can possibly be used. Stainless Steel can be formulated with ‘spring’ characteristics enabling it to self-expand. Nitinol is a metallic alloy of nickel and titanium. It is in a class of metals known as ‘shape memory’. A nitinol-based expansion component can be made with a shape memory of the high-profile expanded state **134**, enabling it to self-expand. There are also shape memory polymers that can be used to construct the expansion component. (163) The braid **124** can be covered with an inner and outer liner to make it atraumatic and prevent arterial wall damage. The inner and outer liners would expand and collapse with the system **100**. (164) The sheath **119** may have an atraumatic tip to aid insertion and eliminate or reduce damage to the artery wall.

(165) The expansion component **110**, sheath **119** or both items could have radio-opaque features so they can be visualized with fluoroscopic imaging.

(166) This embodiment of the system **100** can interface with the other catheterization devices used during the procedure, such as the guide wire **122**, guide catheter **121**, balloon catheters, stent **123**, as well as other medical devices **80**.

VII. HELIX BALLOON EMBODIMENTS

(167) Helix balloon embodiments **104** of the system **100** are similar to tubular balloon embodiments **103** of the system **100**, except that in a helix balloon embodiment **104** of the system **100**, the balloon **111** is constrained and shaped by a matrix **114** the configures the shape of the balloon **111** into a helix balloon **113**.

A. HELIX BALLOON

(168) Just as a tubular balloon **112** can be inflatable, self-inflating, or mechanically expanding, a helix balloon **113** can change operating modes **130** in precisely the same ways using the same technologies and principles of chemistry and physics.

B. MATRIX

(169) A mechanism or configuration of mechanisms that keep the balloon **111** in the shape of a helix balloon **113**. The matrix **114** maintains the helical shape of the helix balloon **113** in all operating modes **130**. The matrix **114** can be implemented in a wide variety of different embodiments, including but not limited to a weave **145**, a bonding agent **146**, a thermally formed connection **147**, a matrix cover **148**, and a flange **149**. The cross sectional shape of the helix balloon **113** can be maintained differently in different operating modes **130**. For example, the cross section of the helix balloon **113** would otherwise be round in an inflated state (high-profile operating mode **134**) and flat in a deflated state (low-profile operating mode **132**). The matrix **114** can maintain the helical shape in both states. The matrix **114** needs both flexibility and strength to properly perform its function.

(170) The matrix **114** can include a medicinal component **126**, a mechanism or configuration of mechanisms that enable medicinal capabilities to the system **100**. The medicinal component **126** may include diagnosis or treatment of a medical condition, or delivery of medicine or nutrient. The matrix **114** may contain vaso-active agents to cause vasoconstriction or vasodilation, depending on what may be required. Such an agent may be transient or longer lasting. Nitric oxide is an example of a vaso-active agent that can dilate a vessel, which would make the vessel bigger (larger diameter) until the agent wears off. The matrix **114** may contain any of the class of drug coatings that prevent intimal hyperplasia. Intimal hyperplasia often is a physiologic response to an angioplasty procedure resulting in restenosis of the treated area, which in layman's terms is a clogged stent **123**.

(171) 1. Weave

(172) A weave **145** can be a configuration of one or more threads **144** that can contain the balloon **111** in the shape of a helix balloon **113**. The weave **145** can use as many or as few threads **144** as desired. In many embodiments, between 10-12 threads **144** uniformly distributed about the helix balloon **113** is a particular desirable configuration. The weave **145** would wrap around the helix balloon **113** as the helix balloon **113** makes consecutive passes of the helical shape.

(173) 2. Bonding Agent

(174) A chemical means to constrain the shape of the helix balloon **113**. The matrix **114** can be made from a bonding agent **146** that is applied to a balloon **111** to secure its shape as a helix balloon **113**. A bonding agent **146** can be used by itself or with other components to maintain the helical shape of the helix balloon **113**. Consecutive passes of the helical shape can be bonded to adjacent passes. A wide variety of bonding agents including but not limited to adhesive glues or silicone can be used as possible bonding agents **146**. The bonding agent **146** may be applied using dip coating techniques.

(175) 3. Thermally Formed Connection

(176) A constraint on the helix balloon **113** that is implemented through the application of heat. A wide range of thermal forming techniques known in the prior art can be used to connect adjacent passes of the helical shape together. The aggregate configuration of thermally formed connections **147** can by itself or in conjunction with other components, constitute the matrix **114**.

(177) 4. Matrix Covering

(178) A matrix cover **148** is a relatively thin sheet or a collection of thin sheets that overlay the balloon **111** to shape it into a helix balloon **113**. The matrix cover **148** can contain the helix balloon **113** and maintain its helical shape. The matrix cover **148** can be made from a fabric or other similar material suitable for the particular location **88** in the patient **90**. The matrix cover **148** can cover a single pass of the helical shape, multiple passes or all passes. The matrix cover **148** can be used by itself or in conjunction with other components to constitute the matrix **114**. The matrix cover **148** may be applied using dip coating techniques as well as other plausible manufacturing methods.

(179) 5. Flange

(180) A flange **149** is a rim, collar, or ring that secures the balloon **111** into the shape of a helix

balloon **113**. The cross-section of the helix balloon **113** can have one or more flanges **149**. Adjacent passes of the helical shape can be connected together by the flange **149**. The connected flanges **149** in the aggregate can form the matrix component **114**. Flanges **149** can be connected using a weave **145**, a bonding agent **146**, a thermally formed connection **147**, a matrix cover **148**, and/or potentially other means.

C. EXAMPLES

(181) FIG. **7a** is a diagram illustrating a perspective view of a helix **113** and matrix **114** configuration that includes a tubular balloon constrained in the shape of a helix by a weave **145** functioning as a matrix **114**. The central lumen **120** inside the helix is 0.058 inches, which is created by wrapping the tubular balloon **112** around a mandrel and secured by the matrix **114**. Twelve threads **144** that are 0.002 inches in diameter form the matrix **114**.

(182) FIG. **7b** is a diagram illustrating an example of a side view of the helix **113** and matrix **114** configuration of FIG. **7a**.

(183) FIG. **7c** is a diagram illustrating an example of a planar front view of the helix **113** and matrix **114** configuration of FIGS. **7a** and **7b**. As illustrated in the figure, the 12 threads are uniformly spaced around the helix balloon **113**.

(184) FIG. **7d** is a diagram illustrating an example of a perspective section view of the helix **113** and matrix **114** configuration of FIGS. **7a-7c**.

(185) FIG. **7e** is a diagram illustrating an example of a close-up view of the illustration in FIG. **7d**.

(186) FIG. **7f** is a hierarchy diagram illustrating various examples of different helix balloon **113** and matrix components **114**. As illustrated by the dotted line in the figure, the matrix **114** is an optional component although often a highly desirable one. As illustrated in the Figure, a helix balloon **113** can be implemented as a self-expanding helix component **141**, a mechanically-expanding helix component **142**, as well as the inflatable helix balloon **113** illustrated in FIGS. **7a-7e**. As illustrated in the Figure, the matrix **114** can be implemented as a weave **145**, a bonding agent **146**, a thermally formed connection **147**, and a matrix cover **148**. As discussed above, the matrix **114** can include a medicinal component **126**.

VIII. GLOSSARY/INDEX

(187) Table 1 below is a chart linking together element numbers, element names, and element descriptions.

(188) TABLE-US-00001 TABLE 1 # Name Description 80 Medical A device that serves a medical purpose within the Device body of the patient 90. The system 100 creates the lumen 120 in order to provide space for the medical device 80 to be positioned at a desired location 88 within the body of the patient 90. 81 Medical A process performed on or in a patient 90 by a Procedure provider 92 for the purpose of benefiting the health status of the patient 90. Examples of medical procedures 81 that can benefit from the creation of a lumen 120 or the enhancement of a lumen 120 can include but are not limited to Percutaneous Coronary Intervention (PCI), Percutaneous Coronary Angiogram (PCA), Chronic Total Occlusions (CTO), Stent implantation, Atherectomy, and Embolic Protection. Although the system 100 was originally devised to assist providers 92 with respect to coronary vascular procedures, the system 100 can benefit patients 90 in other contexts. 88 Desired A position within the body of the patient 90 that Location the provider 92 desires to create a lumen 120 for the insertion of a medical device 80 and/or the performance of a medical procedure 81. 90 Patient The beneficiary of the medical device 80. The patient 90 is the organism in which the lumen 120 is created for the purposes of positioning and utilizing the medical device 80. The system 100 can be used with respect to a wide variety of different types of patients 90 including but not limited to, human beings, other types of mammals, other types of animals, and other living organisms. 91 Blood Vessel A passageway in the body of the patient 90 through which blood circulates. 92 Provider A person who provides health care assistance to the patient 90. The provider 92 is typically a physician 92, but other professionals such as nurses, paramedics, physician assistants, etc. may also act as providers 92 with respect to the system 100. 100 System A

collection of components that collectively provide for the functionality of creating a space 120 within a body. 101 Direct Embodiments of the system 100 that directly Expansion inflate or deflate the expansion component 110 in Embodiments order to change operating modes 130. Direct expansion embodiments 101 can include but are not limited to a balloon 111, such as a tubular balloon embodiments 103 and helix balloon embodiments 104. 102 Indirect Embodiments of the system 100 that utilize other Expansion components of the system 100 to expand or shrink Embodiments the expansion component 110. Indirect expansion embodiments 102 can include but are not limited to guide balloon embodiments 105 (expansion component 110 expands by advancing on a guide balloon 115), insertion component embodiments 106 (expansion component 110 expands by the insertion of an insertion component 117), and sheathed balloon embodiments 107 (expansion component 110 expands when it is removed from and no longer constrained by the sheath 119). 103 Tubular An embodiment of the system 100 where the Balloon expansion component 110 is a tubular balloon Embodiments 104 Helix Balloon An embodiment of the system 100 where the Embodiments expansion component 110 is a helix balloon. 105 Guide Balloon An embodiment of the system 100 where a the Embodiments expansion component 110 is advanced over a guide balloon 115 (which is a type of balloon 111) that is in an inflated state in order to expand the expansion component 110 from a low-profile operating mode 132 into a high-profile operating mode 134. 106 Insertion An embodiment of the system 100 where an Component insertion component is inserted into the expansion Embodiments component 110 to expand the expansion component 110 from a low-profile operating mode 132 into a high-profile operating mode 134. 107 Sheathed An embodiment of the system 100 where a Balloon sheathed balloon 118 is removed from a sheath Embodiments 119 to change from a low-profile operating mode 132 into a high-profile operating mode 134. The sheathed balloon 118 expands when no longer constrained by the sheath 119. 108 Expansion Embodiments of the system 100 that involve Component some type of a balloon 111 as the expansion Balloon component 110. Examples of expansion Embodiments component balloon embodiments 108 can include but are not limited to tubular balloon embodiments 103, helix balloon embodiments 104, and sheath embodiments 107. 109 Expansion Embodiments of the system 100 that do not Component involve an expansion component 110 that is a Non-Balloon balloon 111. Examples of expansion component Embodiments non-balloon embodiments 109 can include but are not limited to guide balloon embodiments 105 (expansion component 110 is advanced onto an inflated guide balloon 115) and insertion component embodiments 106 (insertion component 117 such as a second guide catheter 121 is inserted into the expansion component 110). 110 Expansion Potentially any mechanism that can expand from Component a low-profile operating mode 132 into a high- profile operating mode 134 to create the space 120. 111 Balloon An at least semi-flexible container, such that filling the container changes the shape of the container. Balloons can be inflated with air, other types of gasses, water, and other types of liquids. Some embodiments of balloons 111 can be inflated utilizing mechanical means. Many categories of expansion components 110 are balloons 111 (tubular balloon embodiments 103, helix balloon embodiments 104, and sheathed balloon embodiments 107) or are used in conjunction with balloons 111 (guide balloon embodiments 105). 112 Tubular A balloon 111 with a “donut hole” in the center of Balloon the balloon 111. When the tubular balloon 112 is inflated, the “donut hole” at the center of the balloon 111 is the lumen 120. 113 Helix Balloon A balloon 111 that is helix or helical shaped, like a coil or spring. The center of the helix can be used to create a lumen 120 when the helix balloon 113 expands from a low-profile state 132 into a high-profile state 134. The helix balloon 113 may be coupled with a matrix 114 to reinforce and augment the desired shape and structural attributes of the helix balloon 113. 114 Matrix or A mechanism or configuration of mechanisms Matrix that keep the balloon 111 in the shape of a helix Component balloon 113. The matrix 114 maintains the helical shape of the helix balloon 113 in all operating modes 130. The matrix 114 can be implemented in a wide variety of different embodiments, including but not limited to a weave 145, a bonding agent 146, a thermally

formed connection 147, a cover 148, and a flange 149. The cross sectional shape of the helix balloon 113 can be maintained differently in different operating modes 130. For example, the cross section of the helix balloon 113 would otherwise be round in an inflated state (high-profile operating mode 134) and flat in a deflated state (low-profile operating mode 132). The matrix 114 can maintain the helical shape in both states. The matrix 114 needs the both flexibility and strength to properly perform its function. The matrix 114 can also be referred to as a matrix component 114.

115 Guide The balloon 111 used in conjunction with a cover Balloon 116 to change the cover 116 from a low-profile operating mode 132 into a high-profile operating mode 134. 116 Cover The expansion component 110 can be implemented as a cover 116 to the guide balloon 115 or to the insertion component 117. In the context of an insertion component embodiment 106, the cover 116 can be an integral part of a customary guide catheter 121 in the form of an extension on the distal end of the guide catheter 121. In many such embodiments, the cover 116 can be permanently and irremovably attached from the guide catheter 121 at the time of manufacture. The cover 116 can also be referred to as an expandable cover. 117 Insertion A device that is inserted into the expansion Component component 110 to trigger the expansion of the expansion component 110 from a low-profile operating mode 132 into a high-profile operating mode 134. In some embodiments, the insertion component 117 can be a second guide catheter 121. 118 Sheathed A balloon 111 that is naturally in an expanded Balloon or state. The sheathed balloon 118 changes from a Sheath low-profile operating mode 132 into a high- Balloon profile operating mode 134 when it is removed from a sheath 119. The sheath 119 compresses a sheathed balloon 118 from what would otherwise be a high-profile operating mode 134 into a low- profile operating mode 132. In the some embodiments, the sheathed balloon 118 is a braid 124. 119 Sheath A container of the sheathed balloon 118. The sheath 119 constrains the sheathed balloon 118 such that the sheathed balloon 118 remains in a low-profile operating mode 132 so long as the sheathed balloon 118 remains within the sheath 119. Upon removal from the sheath 119, the sheathed balloon 118 expands from a low-profile operating mode 132 into a high-profile operating mode 134. 120 Lumen Space in the body of the patient 90 that is created by system 100. “Lumen” 120 is a medical term of art. The space is typically in the shape of a passageway or tunnel through the expansion component 110 for use by other medical devices 80 and/or in the performing of medical procedures 81 in the treatment of a patient 90. The transition of the expansion component 110 from a low-profile operating mode 132 into a high-profile operating mode 134 creates a lumen 120. 121 Guide A tube through which other medical devices 80 or Catheter the expansion component 110 and other components of the system 100 can be inserted and positioned within the patient 90. Guide catheters 121 are a very common and fundamental medical device 80 used for vascular catheterization procedures. Different embodiments of the system 100 can involve zero, one, two, or even 3 or more guide catheters 121. 122 Guide Wire A wire or similar cord used to “guide” other medical devices 80 to the desired location 88 within the patient 90. It can also be used to connect different components of the system 100 to each other. It is often useful to have a relatively thin wire 122 act in the lead of other components of the system 100. The guide wire 122 is a very common and fundamental medical device 80 used for vascular catheterization procedures. 123 Stent A type of medical device 80 that can be implanted within the blood vessel 91 of a patient 90 to keep the vessel 91 open for blood flow. Some embodiments of the system 100 are intended to create a lumen to facilitate inserting the stent 123 to the desired location 88. The stent 123 can also be referred to as a stent catheter. 124 Braid or A type of self-expanding sheathed balloon 118 Braid and a type of expansion component 110. The Balloon construction of the braid 124 can be designed to provide optimum performance. Braid 124 characteristics such as number of wires, shape of wire, wire material, pitch, uniform pitch, variable pitch and weave pattern can be chosen to obtain the desired performance. More or less wires, and wire material, can affect strength and flexibility of the component. Round wires or flat wires can affect wall thickness. Pitch and weave pattern can affect expansion strength and profile size. 125 Attachment A wire that is attached to a balloon 111 or other Wire form of

expansion component 110. Unlike a guide wire 122, the expansion component 110 does not move along the wire 125, but is fixed to the wire 125. 126 Medicinal A substance used in diagnosing and/or treating a Component disease, illness, or medical condition in a patient 90. Some embodiments of the matrix 114 can include a medical component 126, typically in the form of a coating on the matrix 114. The matrix 114 may contain vaso-active agents to cause vasoconstriction or vasodilation, depending on what may be required. Such an agent may be transient or longer lasting. Nitric oxide is an example of a vaso-active agent that can dilate a vessel, which would make the vessel bigger (larger diameter) until the agent wears off. The matrix 114 may contain any of the class of drug coatings that prevent intimal hyperplasia. Intimal hyperplasia often is a physiologic response to an angioplasty procedure resulting in restenosis of the treated area, which in layman's terms is a clogged stent 123. 130 Operating A status or state of the expansion component 110. Mode The expansion component 110 includes at least two operating modes 130: (a) a low-profile operating mode 132; and (b) a high-profile operating mode 134. Some embodiments of the system 100 may involve one or more operating modes 130 between the two extremes of a low- profile operating mode 132 and a high-profile operating mode 134. Many embodiments of the expansion component 110 can transform from a high-profile operating mode 134 back into a low- profile operating mode 132 when the lumen 120 is no longer required or desired. The operating mode 130 can also be referred to as a state 130. 132 Low-Profile The operating mode 130 of the expansion Operating component 110 in which the size of the space 120 Mode is not maximized. Can also be referred to as a low-profile state 132. 134 High-Profile The operating mode 130 of the expansion Operating component 110 in which the size of the lumen Mode 120 is maximized. Can also be referred to as a high-profile state 134. 141 Self- A helix balloon 113 that self-expands. In other Expanding words, the natural default state of a self- Helix expanding helix component 141 is a high-profile Component operating mode 134 rather than a low-profile operating mode 132. 142 Mechanically- A helix balloon 113 that utilizes mechanical Expanding means such as springs to “inflate” (i.e. to Helix transition between operating modes 130) rather Component than a gas or liquid. 144 Thread A cord, fiber, wire, ribbon, strip or other strand of material used in a weave 145. 145 Weave A weave 145 can be a configuration of one or more threads 144 that can contain the balloon 111 in the shape of a helix balloon 113. The weave 145 can use as many or as few threads 144 as desired. In many embodiments, between 10-12 threads 144 uniformly distributed about the helix balloon 113 is a particular desirable configuration. The weave 145 would wrap around the helix balloon 113 as the helix balloon 113 makes consecutive passes of the helical shape. 146 Bonding A chemical means to constrain the shape of the Agent helix balloon 113. The matrix 114 can be made from a bonding agent 146 that is applied to a balloon 111 to secure its shape as a helix balloon 113. A bonding agent 146 can be used by itself or with other components to maintain the helical shape of the helix balloon 113. Consecutive passes of the helical shape can be bonded to adjacent passes. A wide variety of bonding agents including but not limited to adhesive glues or silicone can be used as possible bonding agents 146. The bonding agent 146 may be applied using dip coating techniques. 147 Thermally A constraint on the helix balloon 113 that is Formed implemented through the application of heat. A Connection wide range of thermal forming techniques known in the prior art can be used to connect adjacent passes of the helical shape together. The aggregate configuration of thermally formed connections 147 can by itself or in conjunction with other components, constitute the matrix 114. 148 Matrix Cover A relatively thin sheet or a collection of thin sheets that overlay the balloon 111 to shape it into a helix balloon 113. The matrix cover 148, which can also be referred to as a covering 148, can contain the helix balloon 113 and help maintain its helical shape. The matrix cover 148 can be made from a fabric or other similar material suitable for the particular location 88 in the patient 90. The matrix cover 148 can cover a single pass of the helical shape, multiple passes or all passes. The matrix cover 148 can be used by itself or in conjunction with other components to constitute the matrix 114. The matrix 148 may be applied using dip coating

techniques as well as other plausible manufacturing methods. 149 Flange A flange is a rim, collar, or ring that secures the balloon 111 into the shape of a helix balloon 113. The cross-section of the helix balloon 113 can have one or more flanges 149. Adjacent passes of the helical shape can be connected together by the flange 149. The connected flanges 149 in the aggregate can form the matrix component 114. Flanges 149 can be connected using a weave 145, a bonding agent 146, a thermally formed connection 147, a matrix cover 148, and/or potentially other means. 150 Inflation Tube A passageway to the balloon 111, such as a tubular balloon 112 or a helix balloon 113 that is used to inflate the balloon 111 with air or whatever gas or liquid is used to inflate the balloon 111. 151 Valve The connection between the inflation tube 150 and the balloon 111.

Claims

1. A system for creating a lumen, comprising: a tubular balloon, the tubular balloon operable in a low-profile operating mode and a high-profile operating mode; a matrix constraining the tubular balloon in a generally helical shape, the matrix configured to selectively increase the rigidity of the balloon; and wherein the tubular balloon has a first diameter in the low-profile operating mode and a second diameter in the high-profile operating mode, and the second diameter is larger than the first diameter.
 2. The system of claim 1, wherein the matrix comprises a weave, and wherein the weave further comprises at least one thread.
 3. The system of claim 2, wherein the matrix comprises between ten and twelve threads.
 4. The system of claim 3, wherein the between ten and twelve threads are uniformly distributed about the generally helical shape.
 5. The system of claim 2, wherein the at least one thread is 0.002 inches in diameter.
 6. The system of claim 1, wherein the matrix further comprises at least one of a medicinal component and a nutritional component.
 7. The system of claim 6, where in the matrix comprises the medicinal component, and the medicinal component is a vaso-active agent.
 8. The system of claim 6, where in the matrix comprises the medicinal component, and the medicinal component is a drug coating that prevents intimal hyperplasia.
 9. The system of claim 1, wherein the tubular balloon is deflated in the low-profile operating mode and is inflated in the high-profile operating mode.
 10. The system of claim 9, further comprising an inflation tube in fluid communication with the tubular balloon and configured to provide fluid to inflate the tubular balloon into the high-profile operating mode, and a valve configured to control the flow of fluid from the inflation tube to the tubular balloon.
 11. The system of claim 10, wherein the high-profile mode defines a lumen within the generally helical shape, and wherein the inflation tube is situated longitudinally to the lumen and outside of the lumen.
 12. The system of claim 9, wherein a default state of the tubular balloon is the low-profile operating mode.
 13. The system of claim 1, wherein the tubular balloon is self-expanding.
 14. The system of claim 13, wherein a default state of the tubular balloon is the high-profile operating mode.
 15. The system of claim 1, wherein the tubular balloon is configured to be expanded by mechanical means.
 16. The system of claim 1, wherein the matrix is a thermally-formed connection.
 17. The system of claim 1, wherein the matrix is a bonding agent.
 18. The system of claim 1, wherein the matrix is a covering that overlays the tubular balloon.
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